

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025
Computational Neuroscience [NPT_4130]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the role of neuron morphology in determining neural connectivity and function. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Describe how systems of neurons collaborate to produce sensory-motor coordination. (5)
Course Outcome: "CO1" / Cognitive Level: "Understand"
- 2) Discuss the ionic basis of membrane potential and the role of sodium-potassium channels in generating action potentials. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Application"
B) Formulate the Hodgkin-Huxley model and explain how it captures the excitable membrane behavior of a neuron. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Compare the Hodgkin-Huxley model and the Morris-Lecar model in terms of complexity and applications. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Analyse phase-plane trajectories to interpret how changes in system parameters influence stability and bifurcation behavior in neural models. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyse how receptive field models and spike-triggered averaging techniques reveal stimulus-response relationships in neurons and interpret their role in modelling neural activity. (5)
A) Course Outcome: "CO4" / Cognitive Level: "Analysis"
B) Apply spike train distance measures to neural data and demonstrate how they can be used for discrimination between different stimuli. (5)
Course Outcome: "CO4" / Cognitive Level: "Application"
- 5) Analyse how different forms of short-term plasticity influence temporal filtering and interpret their implications for synaptic information processing. (5)

A) Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Evaluate computational models of long-term plasticity and their implications for adaptive neural computation. (5)

Course Outcome: "CO5" / Cognitive Level: "Evaluate"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4130:Computational Neuroscience

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	5
3	3	CO 3	15
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 NPT_4130:Computational Neuroscience

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	2	10(20%)
4	Analysis	5	25(50%)
5	Create	1	5(10%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)